

Assignment Project: Real-Time Weather & Currency Data Fetcher

Objective

Build a Python application that fetches real-time data from public APIs and displays it in a user-friendly format while using modules, JSON, HTTP requests, dates, and built-in libraries.

Features

Your program should have a menu like this:

```
===== Data Fetcher =====
1. Current Weather
2. Currency Exchange Rate
3. Save Result to JSON File
4. View Previous Saved Data
5. Exit
=====
```

Functional Requirements

1. Current Weather

Ask the user:

Enter city name:

Example:

Dhaka

Use an API to fetch:

- City Name
- Temperature

- Humidity
- Wind Speed
- Weather Condition
- Date & Time

Display

----- Weather Report -----

City: Dhaka

Temperature: 31°C

Humidity: 74%

Wind Speed: 12 km/h

Condition: Cloudy

Fetches At: 22-07-2026 10:30 PM

2. Currency Exchange Rate

Ask user

Base Currency:

USD

Target Currency:

BDT

Use an exchange rate API.

Display

1 USD = 122.45 BDT

Fetches At:

22-07-2026 10:31 PM

3. Save Result

Save the latest fetched data into

data.json

Example

```
{  
  "type": "weather",  
  "city": "Dhaka",  
  "temperature": 31,  
  "humidity": 74,  
  "condition": "Cloudy",  
  "time": "2026-07-22 22:30:10"  
}
```

or

```
{  
  "type": "currency",  
  "base": "USD",  
  "target": "BDT",  
  "rate": 122.45,  
  "time": "2026-07-22 22:31:50"  
}
```

4. View Previous Data

Read the JSON file and display

Last Saved Data

Type: Weather

City: Dhaka

Temperature: 31°C

Saved Time:

2026-07-22 22:30

Handle the case if the file doesn't exist.

Suggested Free APIs

Weather API (No API Key Required)

<https://wttr.in/Dhaka?format=j1>

Returns JSON.

Currency API

<https://open.er-api.com/v6/latest/USD>

or

<https://api.exchangerate-api.com/v4/latest/USD>

Program Structure (Single File)

data_fetcher.py

```
|— import modules
|— weather()
|— currency()
|— save_json()
|— view_json()
|— main_menu()
|— program starts here
```

Only **one .py file** should contain the entire program.

Submission Guideline:

> Submit data_fetcher.py file or Github link of your project.